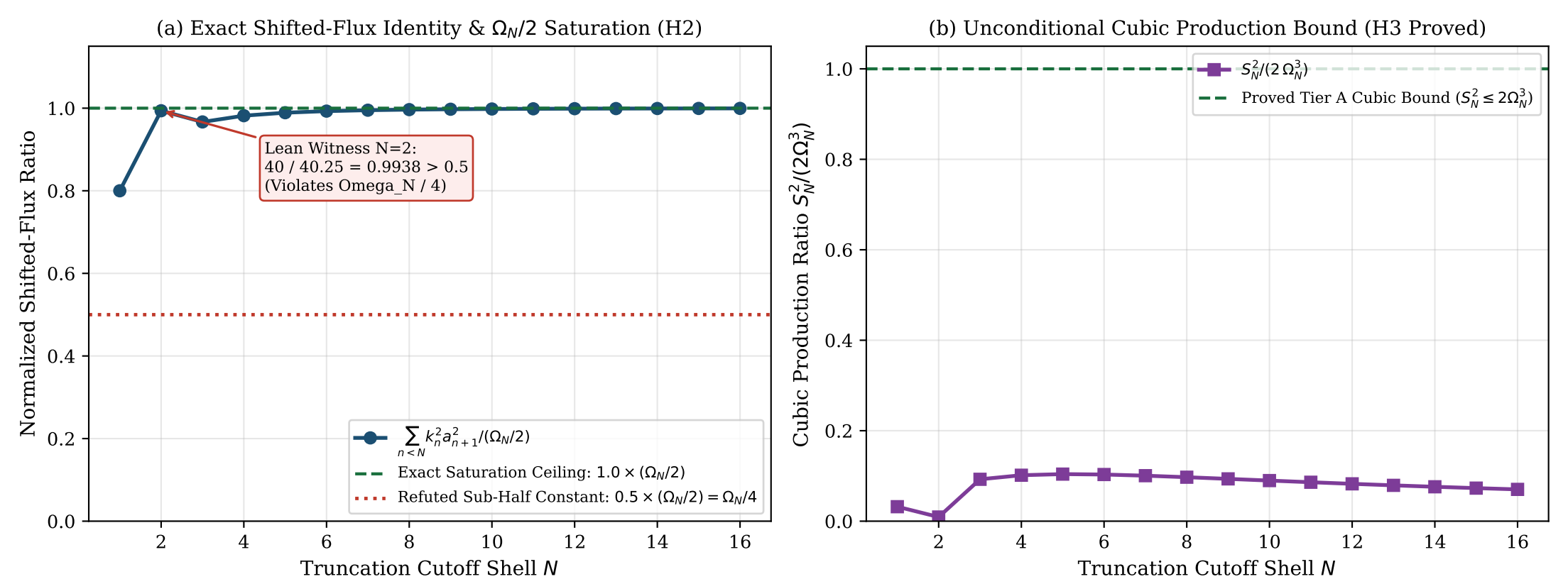
(a) Exact Shifted-Flux Identity & $\Omega_N/2$ Saturation (H2)
Normalized Shifted-Flux Ratio
Truncation Cutoff Shell N
Lean Witness N=2:
40 / 40.25 = 0.9938 > 0.5
(Violates Omega_N / 4)
$\sum_{n<N} k_n^2 a_{n+1}^2/(\Omega_N/2)$
Exact Saturation Ceiling: $1.0 \times (\Omega_N/2)$
Refuted Sub-Half Constant: $0.5 \times (\Omega_N/2) = \Omega_N/4$
(b) Unconditional Cubic Production Bound (H3 Proved)
Cubic Production Ratio $S_N^2/(2\Omega_N^3)$
Truncation Cutoff Shell N
$S_N^2/(2\,\Omega_N^3)$
Proved Tier A Cubic Bound ($S_N^2 \le 2\Omega_N^3$)